APPENDIX

GPT-4 EVALUATOR PROMPTS

To ensure the full reproducibility of our evaluation protocol, we provide the exact prompts used for the GPT-4 evaluator. The evaluation prompt consists of two primary components: **system instructions** (defining the grading criteria) and **few-shot examples** (demonstrating input-output pairs).

Given the diversity of tasks in our benchmark, we categorized the evaluation prompts into two types based on their output format:

- **Classification-based Evaluation (Tasks #2–#7):** Tasks #2 through #7 utilize a consistent qualitative grading scale: `Correct`, `Partially`, `Incorrect`, and `Irrelevant`. Since the system instructions for these tasks are structurally identical—differing only in their specific few-shot examples—we present the prompt for **Task #7 (Style Classification)** as a representative example in Figure 2.
- **Score-based Evaluation (Tasks #1 & #8):** Tasks #1 and #8 employ a numerical scoring system ranging from 0 to 3, tailored to their specific objectives.
  - **Task #1 (Primary Color Extraction):** Evaluates the accuracy of color identification. The specific prompt is shown in Figure 1.
  - **Task #8 (Visual Metaphor):** Assesses the semantic similarity of metaphorical interpretations. The specific prompt is shown in Figure 3.

STATISTICAL ANALYSIS AND RELIABILITY

A. Evaluator Reliability and Human Agreement

To validate the reliability of our automated evaluation protocol, we conducted a rigorous study on 640 human-validated samples (80 samples per task). Table I summarizes the agreement metrics:

- **Human-to-Human Agreement:** Measured using Fleiss' κ among three expert annotators. The overall κ of **0.9085** indicates "Almost Perfect" agreement on the ground truth.
- **Evaluator Reliability:** Measured using Cohen's κ between the GPT-4 evaluator and the human consensus. The overall κ of **0.8852** confirms that the evaluator is a reliable proxy for human judgment.

B. Detailed Confidence Intervals

To provide a comprehensive view of model stability, we report the detailed 95% bootstrap confidence intervals (10,000 resamples) for each specific task in Table II.

C. Per-task Standard Deviation across Answer Positions

Table III provides a detailed breakdown of per-task standard deviations, shedding light on the robustness of models against answer position bias. We observe that certain open-source models exhibit specific preferences; notably, the LLaVA-next series (7B and 8B) shows a pronounced systematic bias towards Option A, which drives high variance in recognition and understanding tasks. In contrast, top-tier models like GPT-4o and Gemini demonstrate superior stability, maintaining consistently low standard deviations across almost all design elements. This indicates that leading models are largely insensitive to answer positioning, effectively mitigating the systematic noise observed in smaller architectures.

Task #1: Primary Color Extraction

SYSTEM INSTRUCTION

Please assess the model's response to the question regarding the identification of theme colors. Compare the model's output to the standard answer that has been provided. Allocate points to the model's output based on its accuracy in identifying the correct theme colors as follows:

- Award **1 points** for each theme color that the model correctly identifies and that matches a color in the standard answer.
- The total score for the model's output will range from **0** (if it fails to correctly identify any of the theme colors) to a maximum of **3** (if it successfully identifies three theme colors listed in the standard answer).

Please tally the points accordingly and provide the final score based on the model's output.

FEW-SHOT EXAMPLES

```
<question> Identify and list three primary colors present in the image. Answer:
<gt_answer> The primary colors are:red-purple,white,black.
<model_answer> The main color are rose, yellow, orange.
```

→ <evaluate_output> 0

```
<question> Identify and list three primary colors present in the image. Answer:
<gt_answer> The primary colors are:white,yellow,blue,red.
<model_answer> Green, yellow, orange, black.
```

→ <evaluate_output> 1

```
<question> Identify and list three primary colors present in the image. Answer:
<gt_answer> The primary colors are:white,black,red,red-purple.
<model_answer> Gray, white, yellow, black.
```

→ <evaluate_output> 2

```
<question> Identify and list three primary colors present in the image. Answer:
<gt_answer> The primary colors are:white,black,yellow-orange,red,blue,blue-green.
<model_answer> The theme color for this image are white, blue, red.
```

→ <evaluate_output> 3

Fig. 1: System instruction and few-shot examples for GPT-4 evaluator in Task #1 (Primary Color Extraction).

Task #7: Style Classification

SYSTEM INSTRUCTION

Instructions:
Evaluate the model's output in response to style classification questions by comparing it to the standard answer provided. Assign a grade based on the accuracy of the style identification in the output. Use the following grading scale:

- **Correct:** The style identified in the model's output matches the standard answer perfectly.
- **Partially:** This grade applies in two cases:
  1. The style is correctly identified, but the model's answer to the question is wrong.
  2. The style is incorrectly identified, but the elements of the style are the same as those in the standard answer.
- **Incorrect:** The model's output incorrectly identifies the style of the overall design and the elements within.
- **Irrelevant:** The model's output does not address the question of style at all.

FEW-SHOT EXAMPLES (INPUT → EVALUATOR OUTPUT)

```
<question> Choose the one that is closest to the poster style among the following styles:  vintage,
futuristic, fashion, elegant. Answer:
<gt_answer> The poster style is closest to the vintage style.
<model_answer> The poster style closest to the one in the image is a combination of futuristic and
vintage. It features a mix of geometric shapes and archaic lettering, which is characteristic of
both styles.
```

→ <evaluate_output> Partially

```
<question> Choose the one that is closer to the poster style among the following styles:  artistic,
vintage. Answer:
<gt_answer> The poster is closer to the vintage style.
<model_answer> The image is a pink background with the words 'tangled' written in white letters.
```

→ <evaluate_output> Irrelevant

```
<question> Choose the one that is closer to the poster style among the following styles:  artistic,
vintage. Answer:
<gt_answer> The poster is closer to the artistic style.
<model_answer> The image you provided is in the artistic style. It appears to be a painting or
drawing of a street scene with people and buildings in the background.
```

→ <evaluate_output> Correct

```
<question> Choose the one that is closest to the poster style among the following styles:
creative, futuristic, fashion, elegant,. Answer:
<gt_answer> The poster style is closest to the creative style.
<model_answer> The style of the poster is elegant.
```

→ <evaluate_output> Incorrect

Fig. 2: System instruction and few-shot examples for GPT-4 evaluator in Task #7 (Style Classification).

Task #8: Visual Metaphor

SYSTEM INSTRUCTION

Given the ground truth answer and the model-generated answer below, evaluate the similarity of the metaphorical meaning expressed by both answers. Assign a score from 0 to 3 based on the following criteria:

- **0**: The model-generated answer is irrelevant to the ground truth metaphorical meaning.
- **1**: The model-generated answer is relevant but incomplete or incorrect in its metaphorical representation.
- **2**: The model-generated answer contains one element (either the caption or the metaphor) that is mainly correct, while the other element is relevant.
- **3**: The model-generated answer correctly captures both the caption and the metaphor, aligning closely with the ground truth metaphorical meaning.

FEW-SHOT EXAMPLES

<question> Describe the metaphorical meaning of the image. Answer:
<gt_answer> The metaphorical meaning of the image is: coffee cup is electric switch, caffeine turns on people.
<model_answer> The image of a small cup of coffee with the words 'min' and 'max' written above it can be interpreted as a metaphor for the idea of finding balance or moderation in life. The cup represents a small portion or a limited amount, while the words 'min' and 'max' suggest the extremes or boundaries. The image may encourage viewers to find a balance between these extremes, reminding them that sometimes, less can be more, and moderation is key to achieving a harmonious and fulfilling life.

→ <evaluate_output> 1

<question> Describe the metaphorical meaning of the image. Answer:
<gt_answer> The metaphorical meaning of the image is: cooking spoons and spatulas are car pedals, audi car is a tool that provides taste to food.
<model_answer> The image of three graters hanging on a wall in a dark room can be interpreted metaphorically as a representation of the process of refining or enhancing one's skills or abilities. The graters, which are used to grate or shred ingredients, can symbolize the act of refining or improving something. The dark room might represent the challenges or difficulties that one may face during this process. The fact that the graters are hanging on the wall could signify that they are readily available and prepared for use, emphasizing the importance of continuous practice and dedication to improve one's skills or abilities.

→ <evaluate_output> 0

<question> Describe the metaphorical meaning of the image. Answer:
<gt_answer> The metaphorical meaning of the image is: weapons are family members in a family tree, violence is a transferrable trait inherited from violence.
<model_answer> The image of a gun tree with many guns as its branches and bullets as its leaves represents a metaphorical representation of the proliferation and spread of gun violence in society. The tree, which is a symbol of growth and life, is depicted as being overwhelmed by the sheer number of guns and bullets, suggesting that the issue of gun violence has become a significant and pervasive problem. The image serves as a visual representation of the consequences of the widespread availability and use of firearms, highlighting the need for addressing the root causes of gun violence and promoting safer communities.

→ <evaluate_output> 2

<question> Describe the metaphorical meaning of the image. Answer:
<gt_answer> The metaphorical meaning of the image is: cigarette tip is maze, smoking is a difficult situation to escape. (States of mind are physical locations; obstacles are physical barriers.)
<model_answer> The image shows a cigarette with two people inside it, walking through a maze made by of tobacco leaves. The metaphorical meaning is that quitting smoking can feel like being trapped in a maze, but there is a way out with the help of Champix, a medication designed to aid in smoking cessation. The image suggests that quitting smoking is a journey, but with the right support, it is possible to find a way out and achieve freedom from addiction.

→ <evaluate_output> 3

Fig. 3: System instruction and few-shot examples for GPT-4 evaluator in Task #8 (Visual Metaphor).

TABLE I: Detailed statistical analysis of evaluator reliability and human ground truth quality. We report the inter-annotator agreement (Human-to-Human) using Fleiss' κ and the evaluator agreement (Human vs. GPT-4) using Cohen's κ for each task. Interpretation is based on the Landis & Koch (1977) scale.

Task	**Sample**	**Human-to-Human**		**Human vs GPT-4**	
		Fleiss' κ	**Level**	**Cohen's κ**	**Level**
#1 Primary Color Extraction	80	0.8219	Almost Perfect	0.7500	Substantial
#2 Font Extraction	80	0.9660	Almost Perfect	0.8667	Almost Perfect
#3 Negative Space Detection	80	0.9103	Almost Perfect	0.8833	Almost Perfect
#4 Color Palette Meaning	80	0.9110	Almost Perfect	0.9000	Almost Perfect
#5 Font Style	80	0.9333	Almost Perfect	0.9500	Almost Perfect
#6 Visual Importance	80	0.9218	Almost Perfect	0.9167	Almost Perfect
#7 Style Classification	80	0.9107	Almost Perfect	0.9167	Almost Perfect
#8 Visual Metaphor	80	0.7997	Substantial	0.7833	Substantial
Overall	**640**	**0.9085**	**Almost Perfect**	**0.8852**	**Almost Perfect**

TABLE II: Detailed 95% bootstrap confidence intervals (10,000 resamples) for each specific task. The values in each cell represent the Lower Bound-Upper Bound of the accuracy. Please refer to the main Table IV for the mean accuracy values.

Models	**Element**						**Overall Design**	
	Recognition			Understanding				
	#1 Color	#2 Font	#3 Layout	#4 Color	#5 Font	#6 Layout	#7 Style	#8 Metaphor
InternVL-v2(1B)	33.3-42.8	19.6-25.5	13.5-18.5	29.8-36.2	26.4-32.6	20.2-26.0	38.8-45.6	8.8-13.0
LLaVA-OneVision(0.5B)	42.7-50.7	22.4-28.4	23.2-29.4	20.9-26.8	24.2-30.5	21.2-27.3	49.0-55.9	9.3-13.3
LLaVA-next(7B)	54.2-62.7	20.5-26.3	15.4-20.8	22.4-28.5	26.9-33.2	12.4-17.2	50.5-57.4	10.6-15.4
Llama-3.2-Vision(11B)	43.7-53.3	23.5-29.6	19.4-25.1	48.2-55.1	32.6-39.2	19.0-24.8	68.2-74.5	3.4-6.9
LLaVA-next(13B)	61.5-69.2	23.2-29.4	18.2-23.9	48.9-55.9	34.4-41.0	11.6-16.4	76.1-81.8	11.9-17.1
LLaVA-next(8B)	69.2-76.5	27.5-34.0	20.6-26.6	41.8-48.6	39.1-45.9	15.9-21.4	69.9-76.0	11.9-16.5
BLIP-2 (12B)	43.3-53.6	30.2-36.9	26.8-34.2	58.0-65.7	37.3-45.3	27.3-34.7	69.0-76.2	4.1-8.0
Phi-3.5-vision (4B)	60.3-68.3	23.4-29.5	23.4-29.4	54.9-61.8	40.0-46.9	25.2-31.5	83.1-88.0	15.1-20.3
LLaVA-next (34B)	54.3-62.8	37.2-43.9	21.4-27.4	62.3-68.8	40.0-46.9	25.2-31.5	79.2-84.5	14.5-19.8
Janus-Pro (7B)	62.3-70.8	27.6-34.0	19.6-25.5	60.6-67.2	44.9-51.7	28.6-35.1	76.5-82.0	14.4-19.9
Mistral-small-3.1(24B)	46.5-55.5	41.4-48.2	36.0-42.8	77.5-83.0	54.8-61.5	25.5-31.8	83.0-87.9	10.6-15.7
InternVL-v2 (40B)	69.2-76.5	42.2-49.2	23.5-29.6	76.8-82.2	51.5-58.2	27.3-33.5	84.6-89.2	18.4-24.2
Gemma-3(27B)	56.8-65.3	37.1-44.0	48.2-55.0	80.0-85.2	60.1-66.9	26.4-32.8	85.6-90.2	14.3-20.3
InternVL-v2.5 (78B)	65.8-73.8	42.6-49.6	32.5-39.1	80.0-85.2	60.9-67.6	28.9-35.2	84.0-88.8	18.8-25.0
Gemini-1.5	42.3-50.5	51.7-58.8	50.2-57.1	80.9-85.9	63.5-70.0	26.9-33.2	87.5-91.8	25.6-33.0
Qwen-VL-Max	71.5-79.0	40.0-46.9	58.1-64.9	80.6-85.9	51.0-57.8	26.9-33.2	85.9-90.4	15.8-21.8
Gemini-2	48.8-58.2	54.2-61.0	64.5-71.0	83.5-88.4	67.0-73.2	28.9-35.2	86.9-91.2	28.1-35.5
GPT-4o	67.8-75.2	65.8-72.1	55.4-62.1	90.4-94.1	69.0-75.1	32.6-39.1	88.6-92.6	29.6-36.7
GPT-4.1	57.5-66.0	68.2-74.6	63.4-69.9	91.6-95.0	69.6-75.8	24.5-30.8	89.4-93.2	34.4-43.1

TABLE III: Per-task Standard Deviation (Std) across answer positions (A, B, C, D). Tasks #1 (Color Theme) and #8 (Visual Metaphor) are excluded as they are not multiple-choice. To analyze position bias, values represent the standard deviation of accuracy when the correct answer is placed at different positions. **Std values exceeding 5%** are annotated with a subscript (e.g., 12.5_{C}), indicating the specific option position (A–D) that most skewed the result (i.e., showed the largest deviation from the mean).

Models	**Element**					**Overall Design**
	Recognition		Understanding			
	#2 Font	#3 Layout	#4 Color	#5 Font	#6 Layout	#7 Style
InternVL-v2(1B)	1.5	1.5	7.9_{A}	6.0_{C}	4.1	13.8_{A}
LLaVA-OneVision(0.5B)	8.2_{A}	5.2_{D}	3.0	3.8	4.0	1.6
LLaVA-next(7B)	14.7_{A}	5.2_{A}	7.6_{A}	8.7_{A}	3.6	3.5
Llama-3.2-Vision(11B)	10.1_{D}	7.7_{A}	8.4_{D}	9.9_{D}	2.2	5.3_{B}
LLaVA-next(13B)	4.8	5.4_{A}	2.1	2.2	3.1	8.3_{A}
LLaVA-next(8B)	13.1_{A}	3.8	6.4_{A}	5.4_{A}	2.8	1.8
BLIP-2 (12B)	2.7	12.3_{D}	3.6	5.3_{A}	18.3_{D}	3.2
Phi-3.5-vision (4B)	5.0_{D}	3.7	6.8_{D}	3.9	1.5	1.9
LLaVA-next (34B)	19.7_{B}	2.6	2.6	5.1_{C}	2.9	2.2
Janus-Pro (7B)	8.4_{A}	3.0	6.6_{A}	21.8_{A}	5.9_{C}	7.1_{C}
Mistral-small-3.1(24B)	4.8	3.5	1.7	3.4	4.1	0.6
InternVL-v2 (40B)	3.5	2.4	1.5	3.8	1.3	1.5
Gemma-3(27B)	12.3_{A}	4.1	2.8	4.9	0.5	3.5
InternVL-v2.5 (78B)	4.9	3.1	4.2	5.7_{A}	3.9	4.1
Gemini-1.5	7.0_{C}	5.7_{A}	1.2	2.2	3.4	2.4
Qwen-VL-Max	3.5	3.5	3.8	2.4	4.0	2.1
Gemini-2	1.9	5.6_{A}	1.9	1.3	2.3	2.4
GPT-4o	2.1	5.9_{A}	0.8	1.8	2.8	0.7
GPT-4.1	3.0	5.2_{A}	0.4	4.3	2.6	2.3